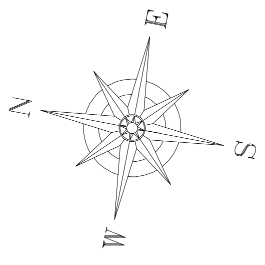
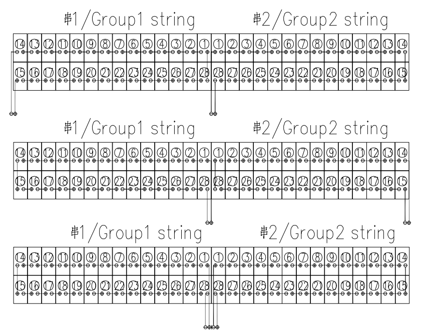


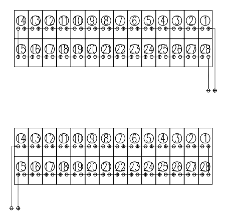


A阵列三种接线方案示意图

Schematic diagram of three wiring schemes for A array

（含有2个组串，56块电池组件）

(Contains 2 string, 56 modules)



B阵列两种接线方案示意图

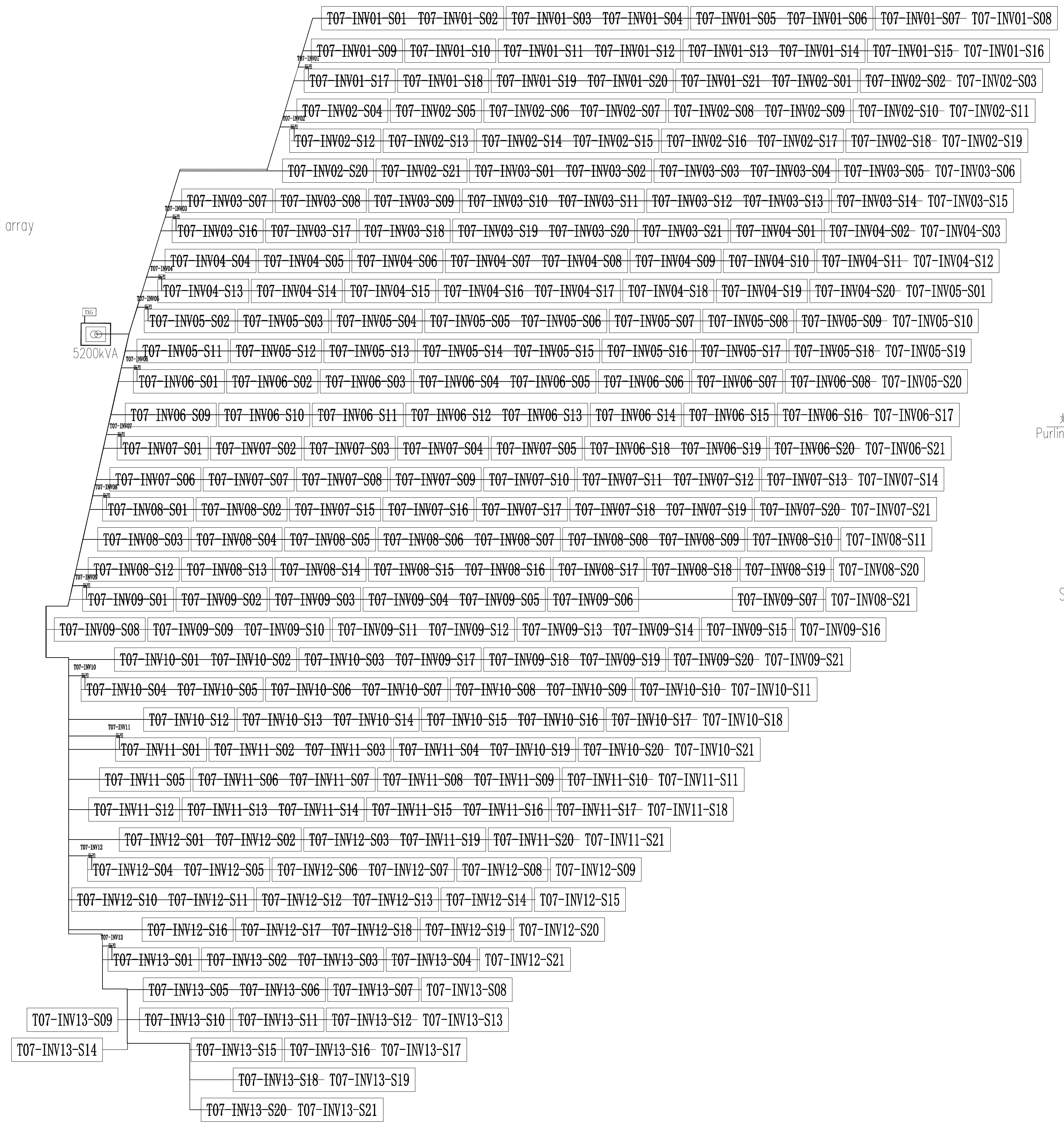
Schematic diagram of three wiring schemes for B array

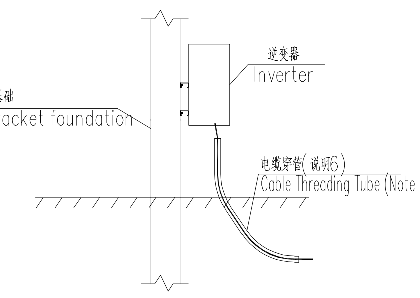
（含有1个组串，28块电池组件）

(Contains 1 string, 28 modules)

图例/Legend：

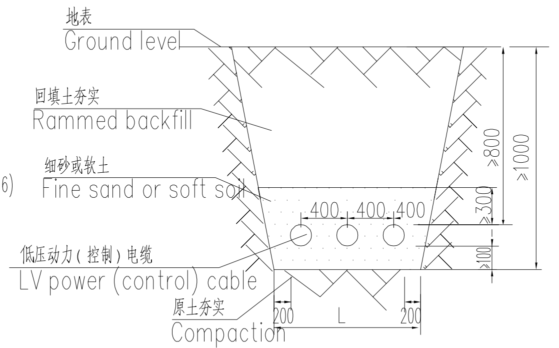
- 组件串联电缆/PV cable
- 二级汇流电缆/Cable connecting inverters and the box transformer
- 箱变/Box Transformer
- 逆变器/Inverter
- TXG 通信柜/Communication cabinet
- B阵列/B array
- A阵列/A array


TXG
5200kVA
T07-INV01
T07-INV02
T07-INV03
T07-INV04
T07-INV05
T07-INV06
T07-INV07
T07-INV08
T07-INV09
T07-INV10
T07-INV11
T07-INV12
T07-INV13
T07-INV01-S01 T07-INV01-S02 T07-INV01-S03 T07-INV01-S04 T07-INV01-S05 T07-INV01-S06 T07-INV01-S07 T07-INV01-S08
T07-INV01-S09 T07-INV01-S10 T07-INV01-S11 T07-INV01-S12 T07-INV01-S13 T07-INV01-S14 T07-INV01-S15 T07-INV01-S16
T07-INV01-S17 T07-INV01-S18 T07-INV01-S19 T07-INV01-S20 T07-INV01-S21 T07-INV02-S01 T07-INV02-S02 T07-INV02-S03
T07-INV02-S04 T07-INV02-S05 T07-INV02-S06 T07-INV02-S07 T07-INV02-S08 T07-INV02-S09 T07-INV02-S10 T07-INV02-S11
T07-INV02-S12 T07-INV02-S13 T07-INV02-S14 T07-INV02-S15 T07-INV02-S16 T07-INV02-S17 T07-INV02-S18 T07-INV02-S19
T07-INV02-S20 T07-INV02-S21 T07-INV03-S01 T07-INV03-S02 T07-INV03-S03 T07-INV03-S04 T07-INV03-S05 T07-INV03-S06
T07-INV03-S07 T07-INV03-S08 T07-INV03-S09 T07-INV03-S10 T07-INV03-S11 T07-INV03-S12 T07-INV03-S13 T07-INV03-S14 T07-INV03-S15
T07-INV03-S16 T07-INV03-S17 T07-INV03-S18 T07-INV03-S19 T07-INV03-S20 T07-INV03-S21 T07-INV04-S01 T07-INV04-S02 T07-INV04-S03
T07-INV04-S04 T07-INV04-S05 T07-INV04-S06 T07-INV04-S07 T07-INV04-S08 T07-INV04-S09 T07-INV04-S10 T07-INV04-S11 T07-INV04-S12
T07-INV04-S13 T07-INV04-S14 T07-INV04-S15 T07-INV04-S16 T07-INV04-S17 T07-INV04-S18 T07-INV04-S19 T07-INV04-S20 T07-INV05-S01
T07-INV05-S02 T07-INV05-S03 T07-INV05-S04 T07-INV05-S05 T07-INV05-S06 T07-INV05-S07 T07-INV05-S08 T07-INV05-S09 T07-INV05-S10
T07-INV05-S11 T07-INV05-S12 T07-INV05-S13 T07-INV05-S14 T07-INV05-S15 T07-INV05-S16 T07-INV05-S17 T07-INV05-S18 T07-INV05-S19
T07-INV06-S01 T07-INV06-S02 T07-INV06-S03 T07-INV06-S04 T07-INV06-S05 T07-INV06-S06 T07-INV06-S07 T07-INV06-S08 T07-INV05-S20
T07-INV06-S09 T07-INV06-S10 T07-INV06-S11 T07-INV06-S12 T07-INV06-S13 T07-INV06-S14 T07-INV06-S15 T07-INV06-S16 T07-INV06-S17
T07-INV07-S01 T07-INV07-S02 T07-INV07-S03 T07-INV07-S04 T07-INV07-S05 T07-INV06-S18 T07-INV06-S19 T07-INV06-S20 T07-INV06-S21
T07-INV07-S06 T07-INV07-S07 T07-INV07-S08 T07-INV07-S09 T07-INV07-S10 T07-INV07-S11 T07-INV07-S12 T07-INV07-S13 T07-INV07-S14
T07-INV08-S01 T07-INV08-S02 T07-INV07-S15 T07-INV07-S16 T07-INV07-S17 T07-INV07-S18 T07-INV07-S19 T07-INV07-S20 T07-INV07-S21
T07-INV08-S03 T07-INV08-S04 T07-INV08-S05 T07-INV08-S06 T07-INV08-S07 T07-INV08-S08 T07-INV08-S09 T07-INV08-S10 T07-INV08-S11
T07-INV08-S12 T07-INV08-S13 T07-INV08-S14 T07-INV08-S15 T07-INV08-S16 T07-INV08-S17 T07-INV08-S18 T07-INV08-S19 T07-INV08-S20
T07-INV09-S01 T07-INV09-S02 T07-INV09-S03 T07-INV09-S04 T07-INV09-S05 T07-INV09-S06 T07-INV09-S07 T07-INV08-S21
T07-INV09-S08 T07-INV09-S09 T07-INV09-S10 T07-INV09-S11 T07-INV09-S12 T07-INV09-S13 T07-INV09-S14 T07-INV09-S15 T07-INV09-S16
T07-INV10-S01 T07-INV10-S02 T07-INV10-S03 T07-INV09-S17 T07-INV09-S18 T07-INV09-S19 T07-INV09-S20 T07-INV09-S21
T07-INV10-S04 T07-INV10-S05 T07-INV10-S06 T07-INV10-S07 T07-INV10-S08 T07-INV10-S09 T07-INV10-S10 T07-INV10-S11
T07-INV10-S12 T07-INV10-S13 T07-INV10-S14 T07-INV10-S15 T07-INV10-S16 T07-INV10-S17 T07-INV10-S18
T07-INV11-S01 T07-INV11-S02 T07-INV11-S03 T07-INV11-S04 T07-INV10-S19 T07-INV10-S20 T07-INV10-S21
T07-INV11-S05 T07-INV11-S06 T07-INV11-S07 T07-INV11-S08 T07-INV11-S09 T07-INV11-S10 T07-INV11-S11
T07-INV11-S12 T07-INV11-S13 T07-INV11-S14 T07-INV11-S15 T07-INV11-S16 T07-INV11-S17 T07-INV11-S18
T07-INV12-S01 T07-INV12-S02 T07-INV12-S03 T07-INV11-S19 T07-INV11-S20 T07-INV11-S21
T07-INV12-S04 T07-INV12-S05 T07-INV12-S06 T07-INV12-S07 T07-INV12-S08 T07-INV12-S09
T07-INV12-S10 T07-INV12-S11 T07-INV12-S12 T07-INV12-S13 T07-INV12-S14 T07-INV12-S15
T07-INV12-S16 T07-INV12-S17 T07-INV12-S18 T07-INV12-S19 T07-INV12-S20
T07-INV13-S01 T07-INV13-S02 T07-INV13-S03 T07-INV13-S04 T07-INV12-S21
T07-INV13-S05 T07-INV13-S06 T07-INV13-S07 T07-INV13-S08
T07-INV13-S09 T07-INV13-S10 T07-INV13-S11 T07-INV13-S12 T07-INV13-S13
T07-INV13-S14 T07-INV13-S15 T07-INV13-S16 T07-INV13-S17
T07-INV13-S18 T07-INV13-S19
T07-INV13-S20 T07-INV13-S21



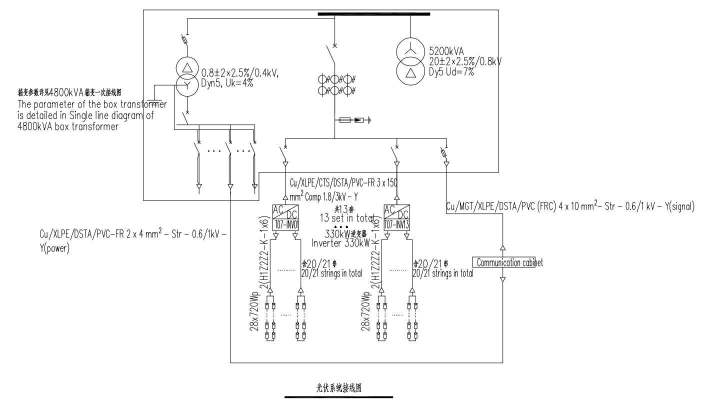
支架基础周围电缆敷设示意图

Schematic diagram of cable laying around purlin bracket foundation



直埋电缆敷设示意图

Directly buried cable laying diagram



光伏系统接线图

PV system wiring diagram

说明/Notes：

1、本图为光伏电场T07发电单元光伏系统接线图，本分区含有13个子分区。本分区共计包含271个光伏组串，7588块电池组件。组件采用单晶硅720Wp带铝边框双面双玻电池组件，尺寸为2384*1303*33mm，重量为37.5kg，由晶科能源股份有限公司供货

1.This is the PV system wiring diagram of the zone T07. Zone T07 contains 13 string inverters, 271 PV strings and 7588 PV modules.

2、光伏阵列：阵列A平面尺寸37.024x4.788m，采用2x28双排竖向布置，含有2个光伏组串；阵列B平面尺寸18.502x4.788m，采用2x14双排竖向布置，含有1个光伏组串；阵列相邻组件间距考虑为20mm

2.Photovoltaic Array: Array A has plane dimensions of 37.024m x 4.788m, arranged in a 2x28 double-row vertical layout, containing 2 photovoltaic strings; Array B has plane dimensions of 18.502m x 4.788m, arranged in a 2x14 double-row vertical layout, containing 1 photovoltaic string. The spacing between adjacent modules in the arrays is set at 20mm.

3、光伏阵列均采用固定式安装，阵列采用朝北安装，安装角度均为6°；南北向相邻组件间的东西向倾角应基本保持一致，角度差控制在1°以内，不应出现明显的交错情况；东西向顺地势布置。

3、All photovoltaic arrays adopt a fixed installation method and are oriented to the north with an installation angle of 6°. The east-west tilt angle between north-south adjacent modules should be essentially consistent, with the angular difference controlled within 1°, and no significant staggering should occur. The arrays are arranged along the terrain in the east-west direction.

4、本分区内13个子分区的组串编号如：分区号—子分区号—组串编号 表示。

4.The string number like "T07-INV01-S01" means：Zone number-Inverter number-string number.

5、本分区共设逆变器13个，逆变器编号与所在分区号—子分区号对应。

5.There are 13 inverters in zone T07, and the inverter number corresponds to the partition number and sub-partition number.

6、一级阵列接线需根据阵列与逆变器的位置合理选择接线方案，尽可能优化组件串联电缆长度。

6.Array wiring should be reasonably selected according to the position of the array and the inverter, and the length of the series cable of the components should be optimized as much as possible.

7、组件1x6电缆优先考虑沿电池组件支架敷设。组件电缆跨支架、下支架、进逆变器或在地面以下敷设时，穿PE管敷设。敷设深度结合现场实际情况，并与业主沟通为准。

7.PV cable is prioritized to be laid along the purlin bracket on PV MMS.When the PV cable crosses the purlin bracket or laid from the purlin bracket to the ground or connected from underground to the inverter or laid underground, it shall be laid through PE pipe. The final depth of the laying needs to be determined based on the actual on-site conditions and communication with the owner.

8、逆变器至箱变的3x150动力电缆需穿Φ100PE管就近敷设到地下800mm进行电缆直埋敷设；地下埋管距地面深度不宜小于0.5m，直埋电缆至地面不得小于0.8m。

8.The power cable with a cross-sectional area of 3x150 from the inverter to the box transformer shall be laid through Φ100 PE pipes to the nearest 800mm underground for direct buried cable laying; The depth of PE pipe buried underground from the ground should not be less than 0.5m, and the directly buried cable to the ground should not be less than 0.8m.

9、电缆过道路或水渠时，穿热镀锌钢管。

9.When the cable passes through the road or canal, it shall wear hot dip galvanized steel pipe.

10、组件电缆数量与保护管截面的关系如下表：

电缆根数	≤4根	≤8根	≤12根	≤20根
PE管管径	PE Φ25	PE Φ50	PE Φ75	PE Φ100

10.The relationship between the quantity of PV cables and the section of the protection tube is as follows：

Quantity of cables	Less than 4	Less than 8	Less than 12	Less than 20
Size of tubes	PE Φ25	PE Φ50	PE Φ75	PE Φ100

序号 No.		对应的电池组件 PV Module				光伏阵列逆变器 Inverter			阵列容量（kWp） Total capacity（kWp）
	发电单元 Installation Zone	串接数（块） Module number of a string	并联数（串） Number of strings	组件数量（块） Number of Modules	组串总数 Total Number of strings	编号 Serial Number	电缆截面 Cable cross section	安装容量（kWp） Installed capacity（kWp）	
1	T07	28	21	588	271	T07-INV01	3x150	423.36	5463.36kWp 单块720Wp 720Wp per module 7588块 7588 modules
2		28	21	588		T07-INV02	3x150	423.36	
3		28	21	588		T07-INV03	3x150	423.36	
4		28	20	560		T07-INV04	3x150	403.20	
5		28	20	560		T07-INV05	3x150	403.20	
6		28	21	588		T07-INV06	3x150	423.36	
7		28	21	588		T07-INV07	3x150	423.36	
8		28	21	588		T07-INV08	3x150	423.36	
9		28	21	588		T07-INV09	3x150	423.36	
10		28	21	588		T07-INV10	3x150	423.36	
11		28	21	588		T07-INV11	3x150	423.36	
12		28	21	588		T07-INV12	3x150	423.36	
13		28	21	588		T07-INV13	3x150	423.36	

设备材料表

EQUIPMENTS & MATERIAL SUMMARISATION SHEET

序号 No.	名称 ITEM	型式及规范 TYPE AND CRITERION	单位 UNIT	数量 QUANTITL	备注 REMARKS
1	组件串联电缆 PV string connection cable	H1Z2Z2-K-1x6mm²	米 m	48000	
2	二级汇流电缆 Cable connecting inverters and the box transformer	Cu/XLPE/CTS/DSTA/PVC-FR 3 x 150 mm² Comp 1.8/3kV - Y	米 m	1000	
3	户外冷缩电缆终端 Outdoor cold shrink cable terminal head	Be applicable to Cu/XLPE/CTS/DSTA/PVC-FR 3 x 150 mm² Comp 1.8/3kV - Y	套 set	26	
4	通信柜信号电缆 Communication cabinet signal cable	Cu/MGT/XLPE/DSTA/PVC (FRC) 4 x 10 mm²- Str - 0.6/1 kV - Y	米 m	10	
5	通信柜电源电缆 Communication cabinet power cable	Cu/XLPE/DSTA/PVC-FR 2 x 4 mm² - Str - 0.6/1kV - Y	米 m	10	
6	网络通信线 Network cable	STP-5e	米 m	10	
7	PE管 PE pipe	Φ25	米 m	120	
8	PE管 PE pipe	Φ50	米 m	100	
9	PE管 PE pipe	Φ75	米 m	80	
10	PE管 PE pipe	Φ100	米 m	150	

GENERAL NOTES

1. ALL DIMENSION ARE IN MILLIMETERS UNLESS OTHERWISE NOTES.
2. ALL ELEVATION ARE IN METERS UNLESS OTHERWISE NOTES.

LEGEND & ABREVIATION

☐	: PV MODULE
🟥	: INVERTER, AC COMBINER & MONITORING PANEL LOCATION
▨	: INTERCONNECTION POINT
▨	: SHADING AREA
▨	: WALKWAY
→∿→	: WATER FLOW
INV	: INVERTER
ACB	: AC COMBINER BOX
MON	: MONITORING PANEL
PYN	: PYRANOMETER
━	: DC CABLE
▭	: AC CABLE
━	: SAFETY LINE
⊡	: LAMP
⊗	: PYRANOMETER

CLIENT APPROVAL

A ☐ APPROVED
B ☐ APPROVED WITH NOTE
C ☐ COMMENTED
D ☐ NOT APPROVED
DATE: SIGN:

REV	DATE	DESCRIPTION	DRW	CHK	RVW	APP	CHK	CHK	APP
			石瀚超	王操	钱庄	贾长杰			
							OE	AHP	AHP

CLIENT :

PT. ARUNA HIJAU POWER

OE:

KWARSA HEXAGON
LAYANAN SOLUSI PEMBANGUNAN UNGGULAN REGIONAL

CONTRACTOR :

中国电建集团湖北工程有限公司
POWERCHINA HUBEI ENGINEERING CO.,LTD

PROJECT TITLE

Pertiwi Lestari Industrial 36MWp
PV Solar Project
XX-XX-XX-XX-XX

DRAWING TITLE

T07发电单元光伏系统接线图
T07 PV System Wiring Diagram

DRAWING NO.

AHP-PL-EL-PDF-PSW-07

CODE STATUS : A. ISSUED FOR REVIEW, B. ISSUED FOR APPROVAL, C. APPROVED FOR CONTRUCTION, D. AS BUILT, CLD. CENCELLED AND INFO = INFORMATION ONLY

SIZE	SHEET	SCALE	CLASS	STATUS	REV.
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